LUIS GUSTAVO TAKANO DE LA CRUZ

FLIGHT DYNAMICS & AVIONICS ENGINEER

CONTACTO

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SOFT SKILLS

Problem solving
Efficient time management

Independent and self-directed
Proactive
Goal-oriented
Teamwork

PROFESSIONAL SKILLS

Electronics
Aerodynamics

Hydraulics
Flight dynamics
Control systems
Statistical modeling
Applied Statistics
Transient analysis

SOFTWARE

Matlab
Simulink
Python
LabView
Arduino IDE
C++
Solidwork

LENGUAGES

Spanish (Native)
English Proficiency: Reading and Writing- Intermediate. Speaking and Listening- Intermediate

PROFESSIONAL ACCOMPLISHMENTS

- Led the development and application of operability models, including control algorithms, to enhance performance and system integration for avionic and aeroderivative systems.
- Programmed and integrated flight control systems and communication protocols, focusing on high-performance standards for hardware-in-the-loop and software-in-the-loop tests.
- Applied statistical optimization and data analysis techniques for transient system identification, aligning with model validation and real-world data integration practices.
- Designed and implemented multivariate architectures for embedded control systems, adhering to standards such as MIL-STD-1797A and NATO (RTO TR-029) regulations.
- Directed model-based control design, ensuring compliance with best practices in control systems design, including thrust command optimization and flight performance analysis.
- Developed tools and frameworks for performance evaluation using programming utilities like Labview to enhance model analysis and streamline development processes.

EDUCATION

2020 - 2023 **PHD IN AERONAUTICAL ENGINEERING**
Center of investigation and innovation in aeronautical engineering, UANL

2017 - 2019 **MASTER'S DEGREE IN AERONAUTICAL**
Center of investigation and innovation in aeronautical engineering, UANL

2010 - 2014 **BACHELOR'S DEGREE IN ELECTRONICS AND AUTOMATION ENGINEERING**
Universidad Autonóma de Nuevo León (UANL),

EXPERIENCE

Professor
UNIVERSIDAD DE MONTERREY (UDEM)
2022 - present

- Introduced practical statistical methods for data analysis, supporting estimation, decision-making, and process forecasting.
- Guided students in understanding data-driven algorithms using matrix operations, vector spaces, and eigenvalues.
- Directed student projects applying scientific methodologies to solve complex engineering challenges with real-world impact.

Research Fellow:
CENTER OF INVESTIGATION AND INNOVATION IN AERONAUTICAL ENGINEERING
2020 - 2023

- Modeled dynamic systems for UAVs, applying multivariate statistical methods for anomaly detection, performance optimization, and operational analysis.
- Published research on **multivariate statistics for anomaly detection**, applying these techniques to turbojet performance analysis, leading to improved model validation and system reliability.
- Integrated **model validation** techniques to align models with real-world operational data, ensuring accuracy in predicting system performance.
- Employed **status-matching models** to compare field data with theoretical predictions, identifying and correcting discrepancies to refine model accuracy.
- Adjusted and optimized **operability models** for gas turbine systems, considering emissions, fuel flexibility, and control aspects, to reflect operational realities and improve system performance.

Research Assistant

CENTER OF INVESTIGATION AND INNOVATION IN AERONAUTICAL ENGINEERING

2017 – 2019

- Designed MIMO control systems for UAVs, focusing on performance, disturbance rejection, and operability within aviation standards (FAR 23/25).
- Conducted software-in-the-loop (SIL) and hardware-in-the-loop (HIL) tests to validate flight controls and optimize operability models before real-world implementation.
- Development of the algorithms of synchronization for the dynamic coupling of multi-agent system
- Aircraft algorithms design for tracking and disturbance rejection in unmanned aerial vehicles in lateral axis through SIL.
- Hardware owner of brushless motor, electronic speed control (ESC), accelerometers and wind tunnel tests

CARTOLITO S.A DE C.V

2015 - 2017

- Corrective and preventive maintenance work on graphic arts machines, fault detection, repair of electronic boards, electrical installations, actuators, electronic relay. Use of temperature, mechanical sensors, inductive sensors, ultrasonic, vision sensors.
- Support and supervised a team of mechanical personnel
- Use and programming of PLC's: Siemens and Mitsubishi, maintenance of AC and DC motors, installation of drives, frequency inverters and temperature controllers.

Social service

INSTITUTE OF BIOTECHNOLOGY

2014 - 2014

- Maintenance of bioprocess units, bioreactors, centrifuges, autoclaves, etc.
- Maintenance of data acquisition card, sensors and bioprocess unit controllers.

Internship

COMISION FEDERAL DE ELECTRICIDAD (CFE)

2013 - 2013

- Preparation of professional practices in the Federal Electricity Commission in the planning department. Used AutoCAD of the medium voltage circuits of the substations in the northern area of the MTY.
- Capture of electrical energy expense, power factor, load factor, etc. of the north zone substations

CERTIFICATIONS

Project Management & Agile Fundamentals (2024)
Yellow Belt (Lean Six Sigma) (2023)
LabxVIEW Associate Developers (CLAD) (2017-2019)

WORSHOPS

- Speaker at Faculty of Mechanical and Electrical Engineering: "Aircraft Flight Dynamics" (July 6th, 2022)
- Speaker at Faculty of Mechanical and Electrical Engineering: "Aircraft Flight Dynamics" (June 28th, 2021)
- Performance of a linear controller for robust stability in a non-linear model of aircraft pitch., L. Takano, J. F. Villarreal, L. A. Amézquita, 2nd International Conference on Aeronautics, Nuevo León, México, 2018
- Designs of and interface based on Matlab-Simulink for the navigation of a quadcopter., L. Takano, L. Feria, L. Cabriales,
- Escobar, O. García, 2nd International Conference on Aeronautics, Nuevo León, México, 2018

PUBLICATIONS AND INDUSTRY PARTNERSHIPS

Formation control as a classical decentralized multivariable problem: Performance, robustness, cross-coupling and perturbation rejection.
Luis G. Takano De La Cruz, Luis Amezquita-Brooks, Octavio Garcia-Salazar, Francisco Villarreal-Valderrama, Carlos Santana- Delgado, Diana Hernandez-Alcantara. Journal of the Franklin Institute (2023), ISSN 0016-0032, https://doi.org/10.1016/j.jfranklin.2023.10.030

Multivariate statistics for anomaly detection: application in a turbojet
Salma Salazar-Martínez, Luis Takano De La Cruz, I. Loboda, Francisco Villarreal-Valderrama, Diana Hernandez-Alcantara, Luis Amézquita-Brooks, journal DYNA, 2023
https://doi.org/10.6036/10921

Design of an aircraft pitch control experimental test bench
F. Villarreal, L. Takano, U. Alvarez, L. Amezquita, E. Liceaga
2018 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC), Ixtapa, Mexico, 2018
doi: 10.1109/ROPEC.2018.8661430

An integral approach for aircraft pitch control and instrumentation in a wind-tunnel
Villarreal Valderrama, J.F., Takano, L., Liceaga-Castro, E., Hernandez-Alcantara, D., Zambrano-Robledo, P.D.C. and Amezquita-Brooks, L. (2020), Aircraft Engineering and Aerospace Technology
https://doi.org/10.1108/AEAT-10-2019-0193